34. The speed of a hydraulic cylinder is controlled by means of a series pressure-compensated flow control valve, as shown in the given circuit. *Given:*

Pump:

$V_g = 25\ \text{cm}^3/\text{rev}$

$n = 1000\ \text{rpm}$

$\eta_v = 0.95$

$\eta_m = 0.93$

$\eta_h = 1$

Directional control valve:

$Q = 6 \times 10^{-7} \sqrt{\Delta P}$

Relief valve:

Cracking pressure = 6 MPa

Override pressure = 0

Hydraulic cylinder:

$f = 2000\ \text{Ns/m}$

$v = 0.1\ \text{m/s}$

$F = 9000\ \text{N}$

Piston diameter = 60 mm

Rod diameter = 25 mm

No inner leakage

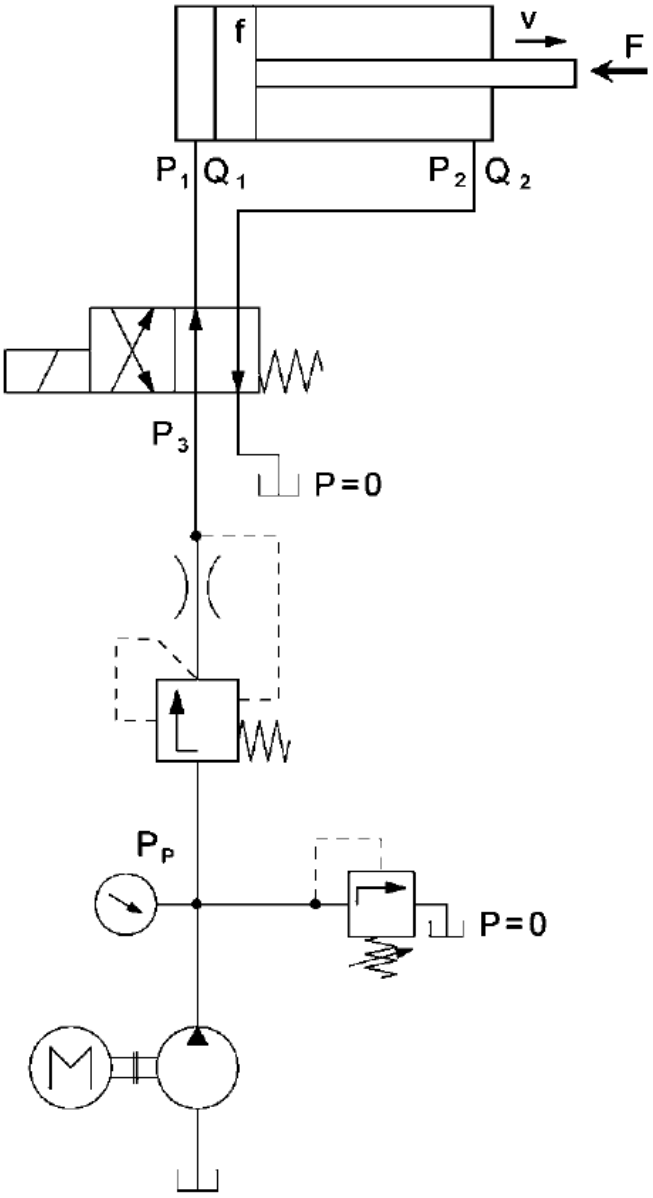
Calculate:

$P_1, P_2, P_3, P_P, Q_1, Q_2$

Power losses in the DCV

Power losses in the FCV

Pump real flow rate



$$Q_t = V_g n = 0.000417 m^3/s$$

$$Q = Q_t \eta_v = 0.000396 m^3/s$$

$$A_p = 0.25\pi D^2 = 0.00283 m^2$$

$$A_r = 0.25\pi(D^2 - d^2) = 0.00234 m^2$$

$$Q_1 = A_p v = 0.000283 m^3/s$$

$$Q_2 = A_r v = 0.000234 m^3/s$$

$$P_2 = \left(\frac{Q_2}{6\times10^{-7}}\right)^2 = 1.52 bar$$

$$P_1 = \left(f * v + F + P_2 A_r\right)/A_p = 33.8 bar$$

$$P_3 = P_1 + \left(\frac{Q_1}{6\times10^{-7}}\right)^2 = 36.012 bar$$

$$N_{DCV} = Q_1(P_3 - P_1) + Q_2 P_2 = 98.2 W$$

$$P_p = 6 MPa$$

$$N_{FCV} = Q_1(P_p - P_3) = 678 W$$

	.000025	Vg	
	16.6666667	n	
	.95	etav	
	.06	D	
	.025	d	
	.1	v	
	2000	f	
	9000	F	
	6000000	Pcr	
Qt=Vg*n		Qt	.000416667
Q=Qt*etav		Q	.000395833
Ap=0.25*pi()*D^2		Ap	.002827433
Ar=0.25*pi()*(D^2-d^2)		Ar	.00233656
Q1=Ap*v		Q1	.000282743
Q2=Ar*v		Q2	.000233656
p2=(Q2/6e-7)^2		p2	151653.068
p1=(f*v+F+p2*Ar)/Ap		p1	3379158.8
p3=p1+(Q1/6e-7)^2		p3	3601224.9
NDCV=Q1*(p3-p1)+Q2*p2		NDCV	98.2223526
NFCV=Q1*(Pcr-p3)		NFCV	678.23768